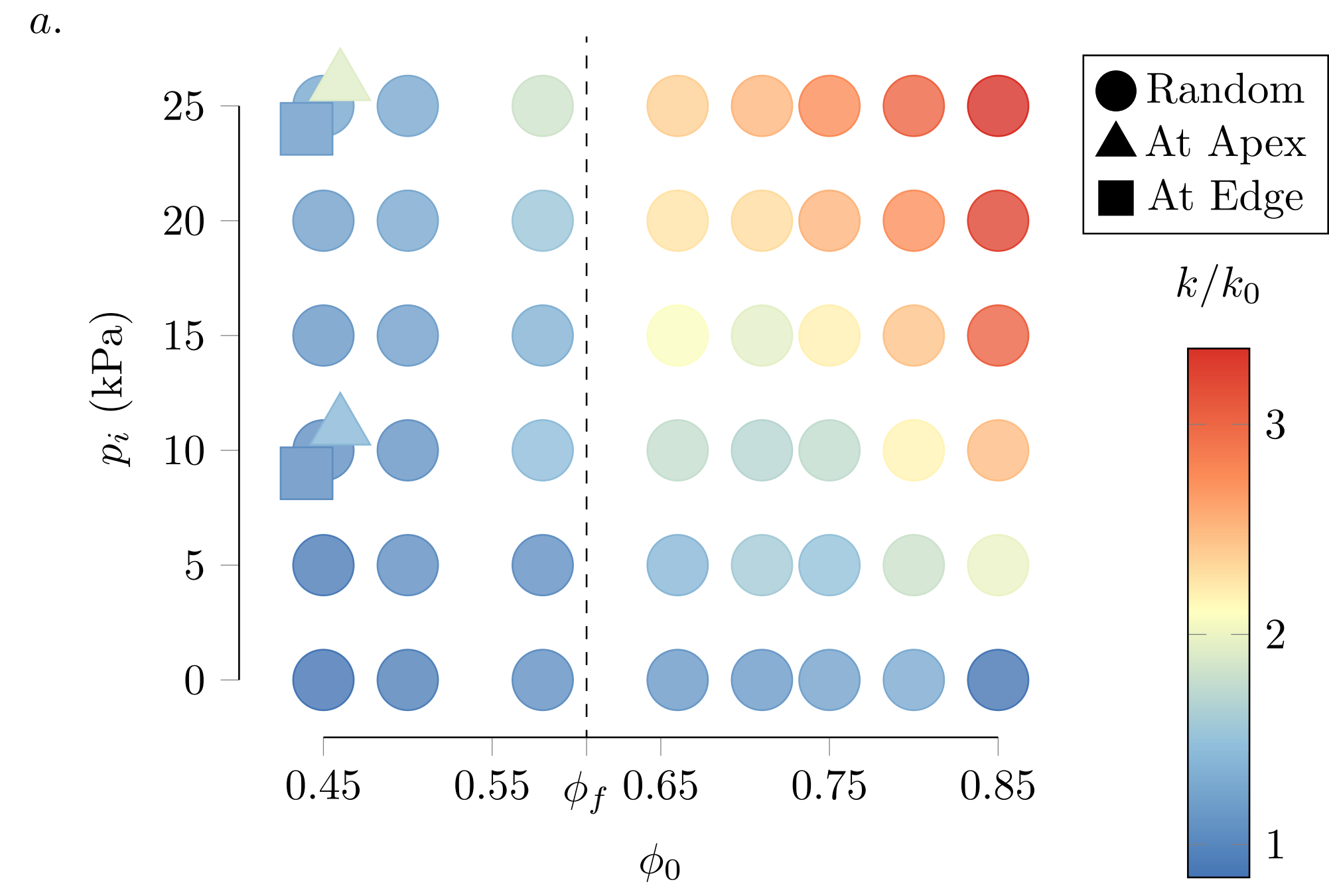
a.
Random
At Apex
At Edge
k/k_0
3
2
1
25
20
15
10
5
0
p_i (kPa)
0.45
0.55
ϕ_f
0.65
0.75
0.85
ϕ_0